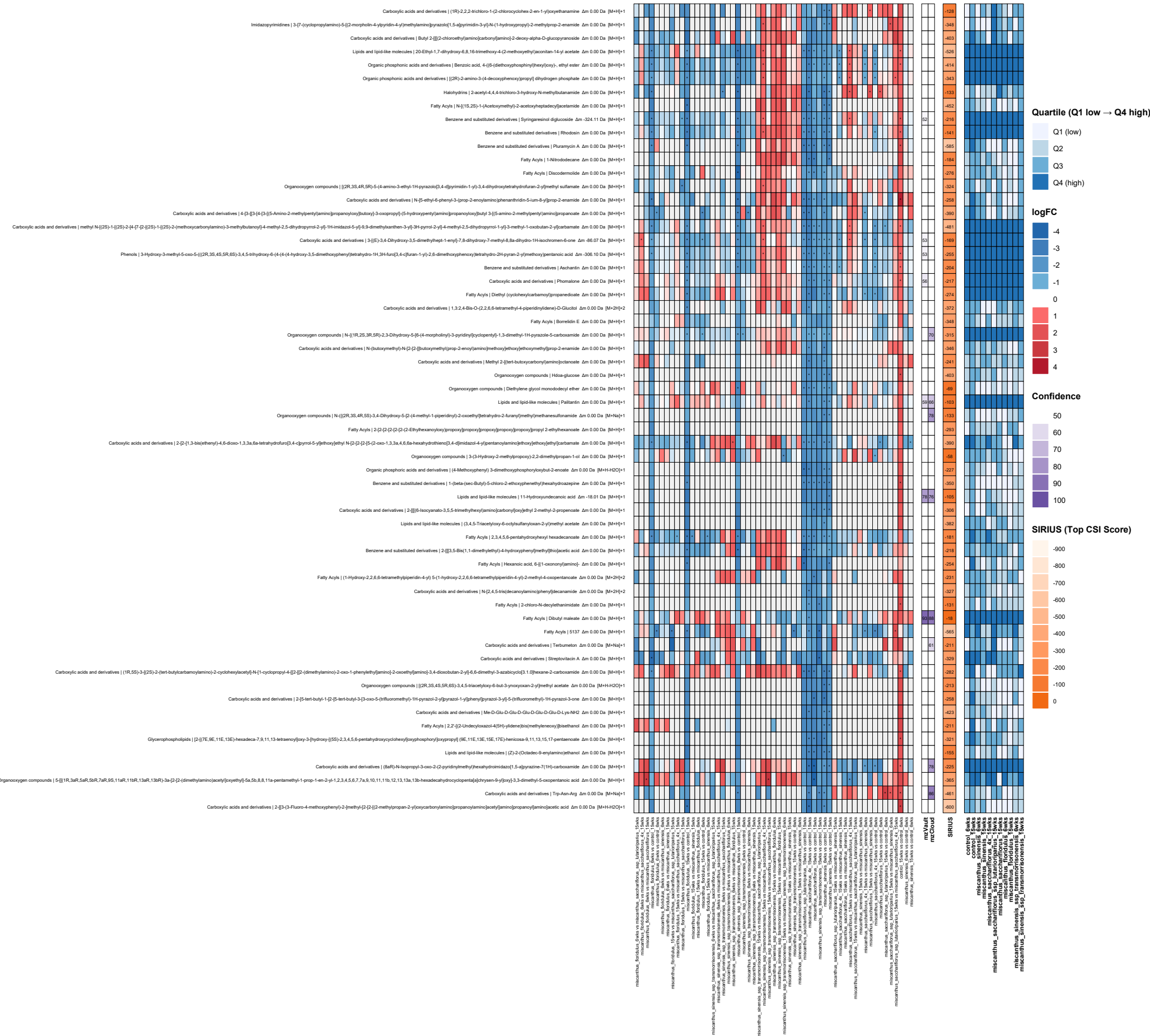
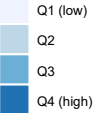


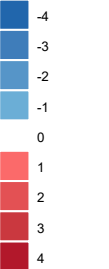




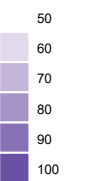
Quartile (Q1 low → Q4 high)



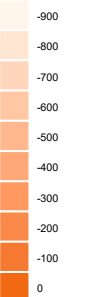
logFC

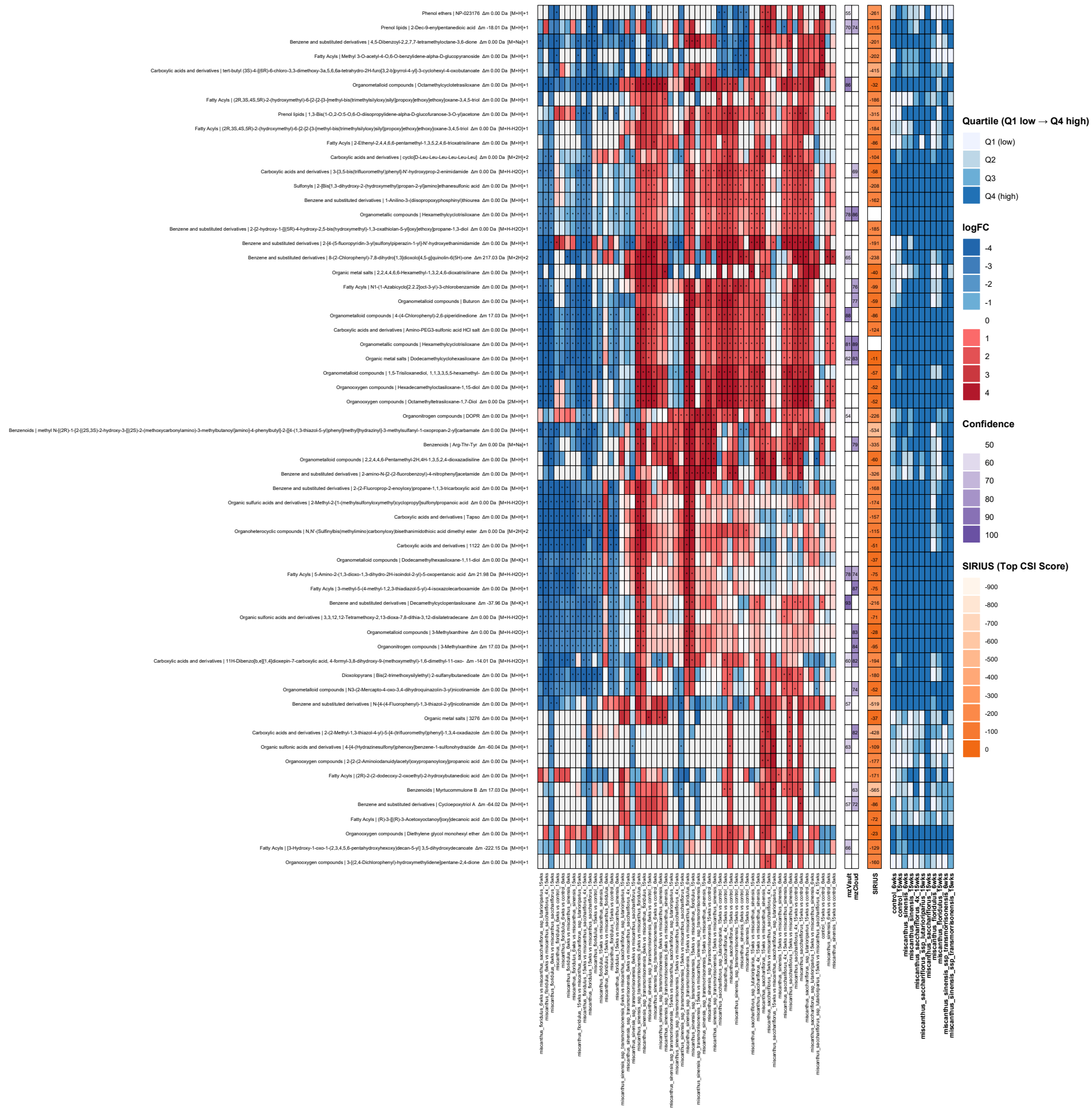


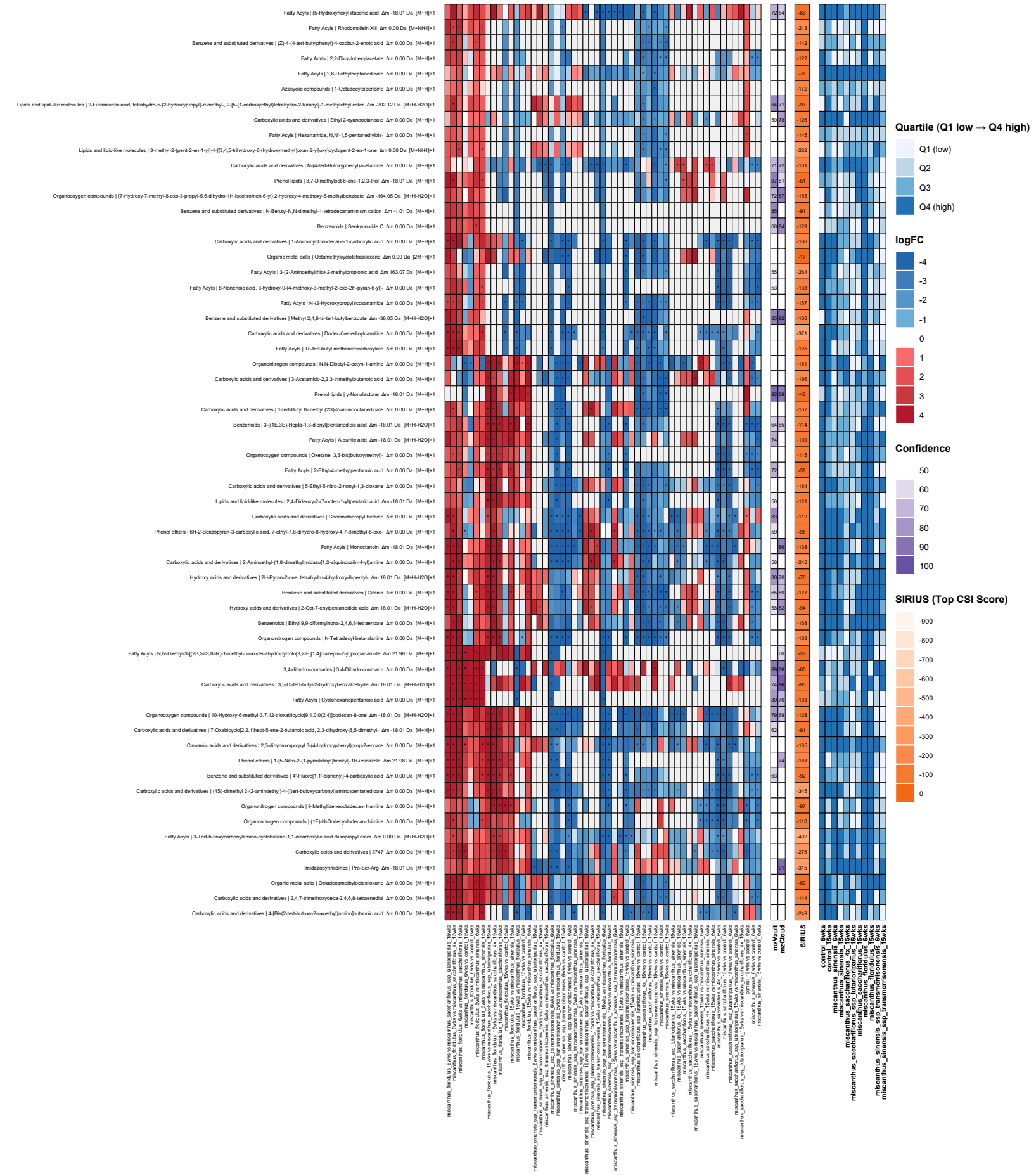
Confidence

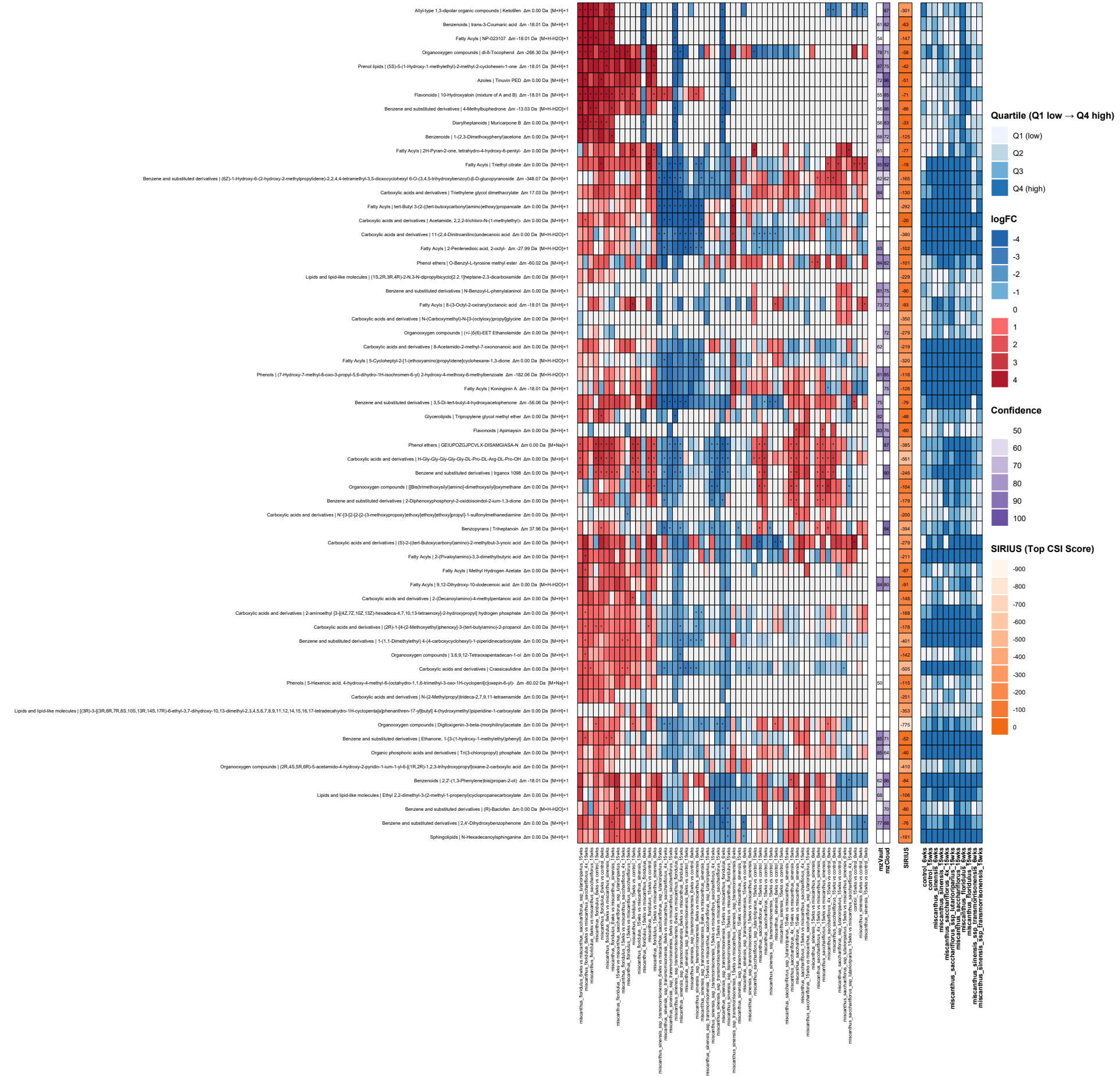


SIRIUS (Top CSI Score)

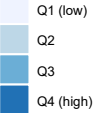




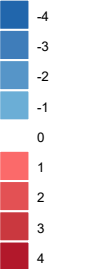




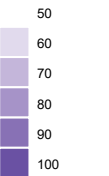
Quartile (Q1 low → Q4 high)



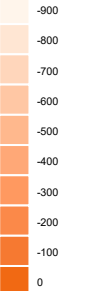
logFC



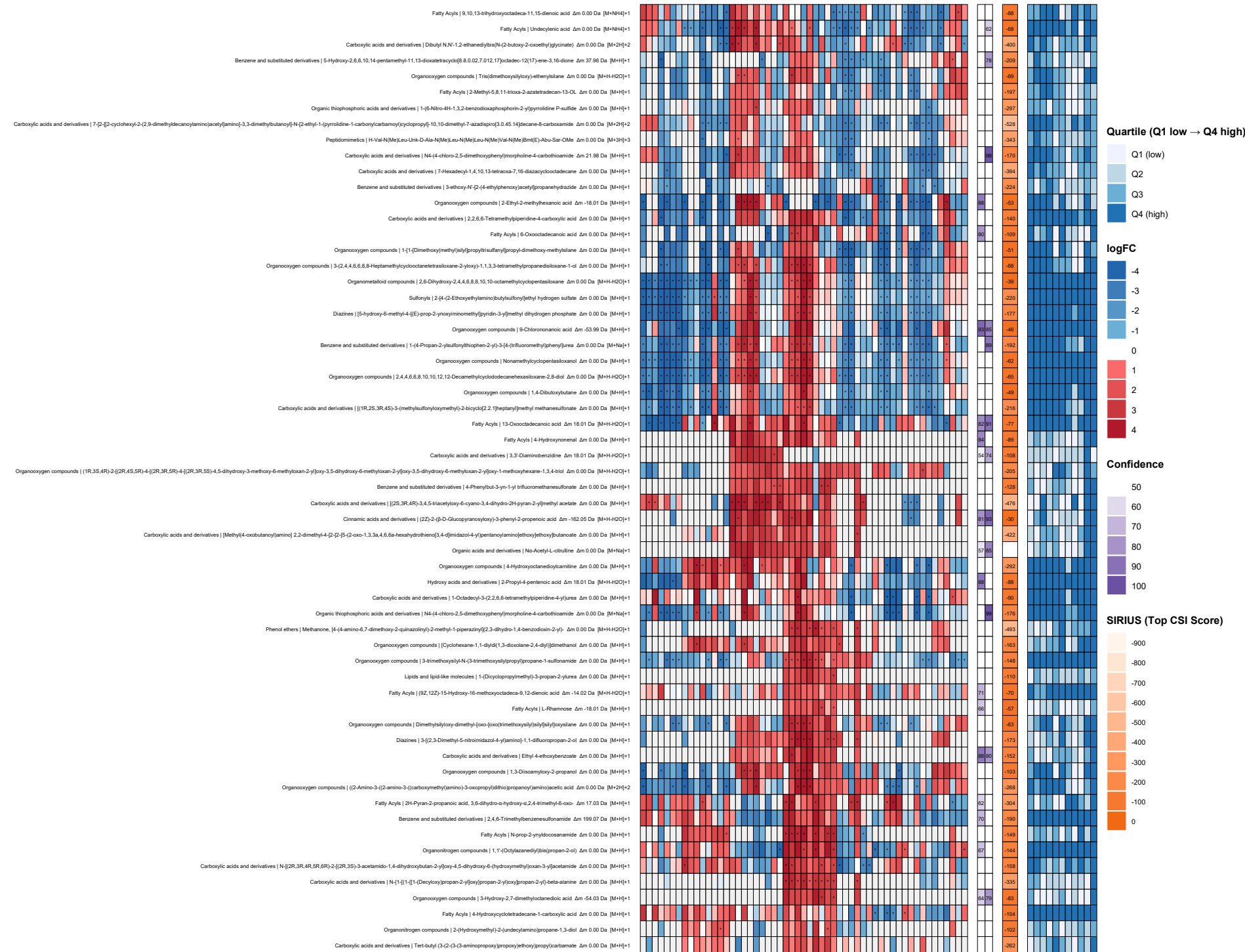
Confidence



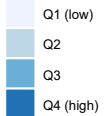
SIRIUS (Top CSI Score)



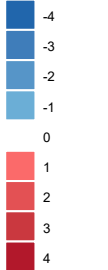




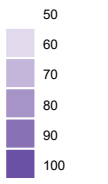
Quartile (Q1 low → Q4 high)



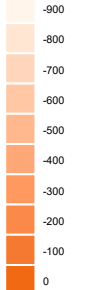
logFC



Confidence



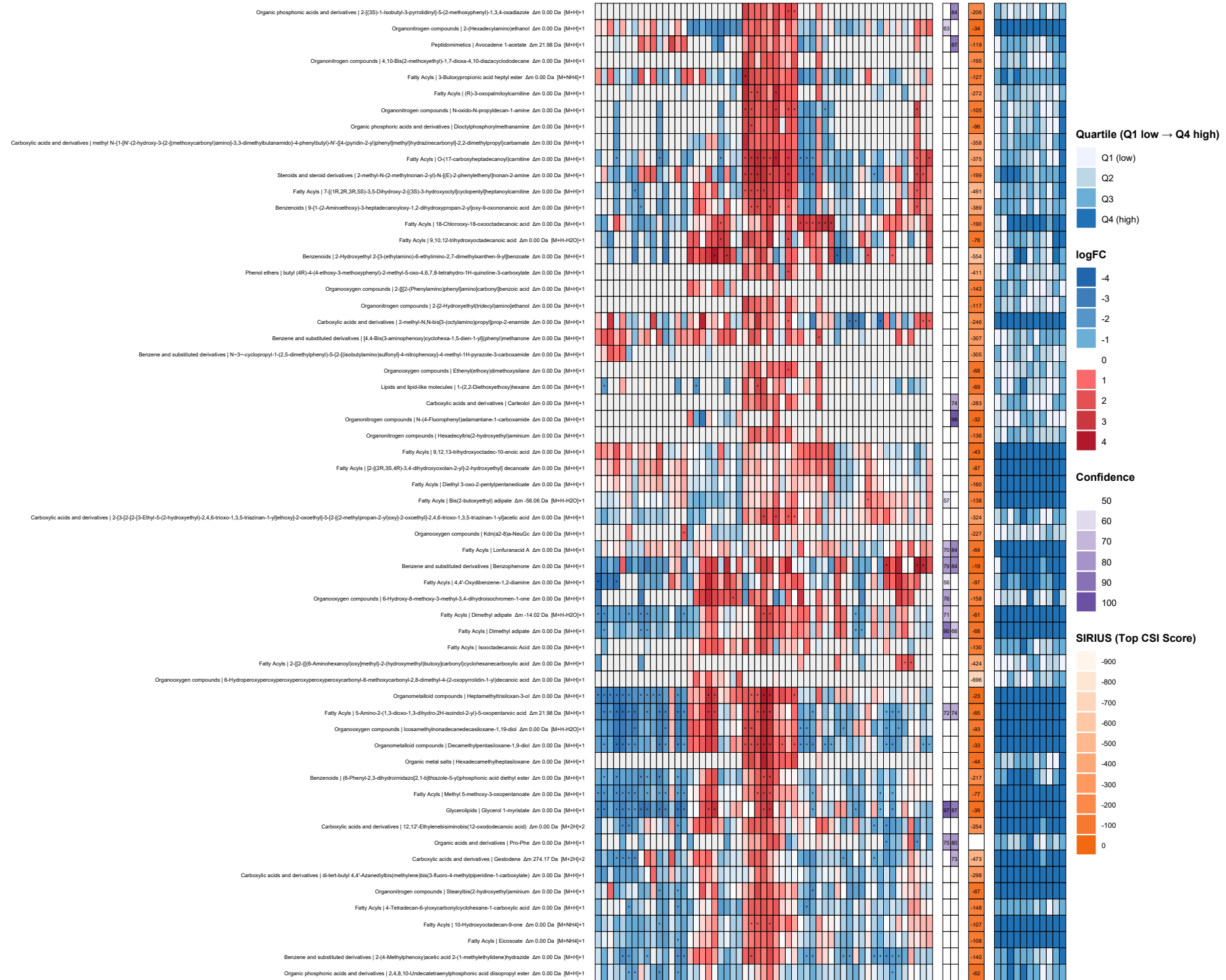
SIRIUS (Top CSI Score)



metabolite names and molecular weights/formulas (MW, MF) are listed on the left side of the heatmap. The columns are labeled with sample IDs (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13).

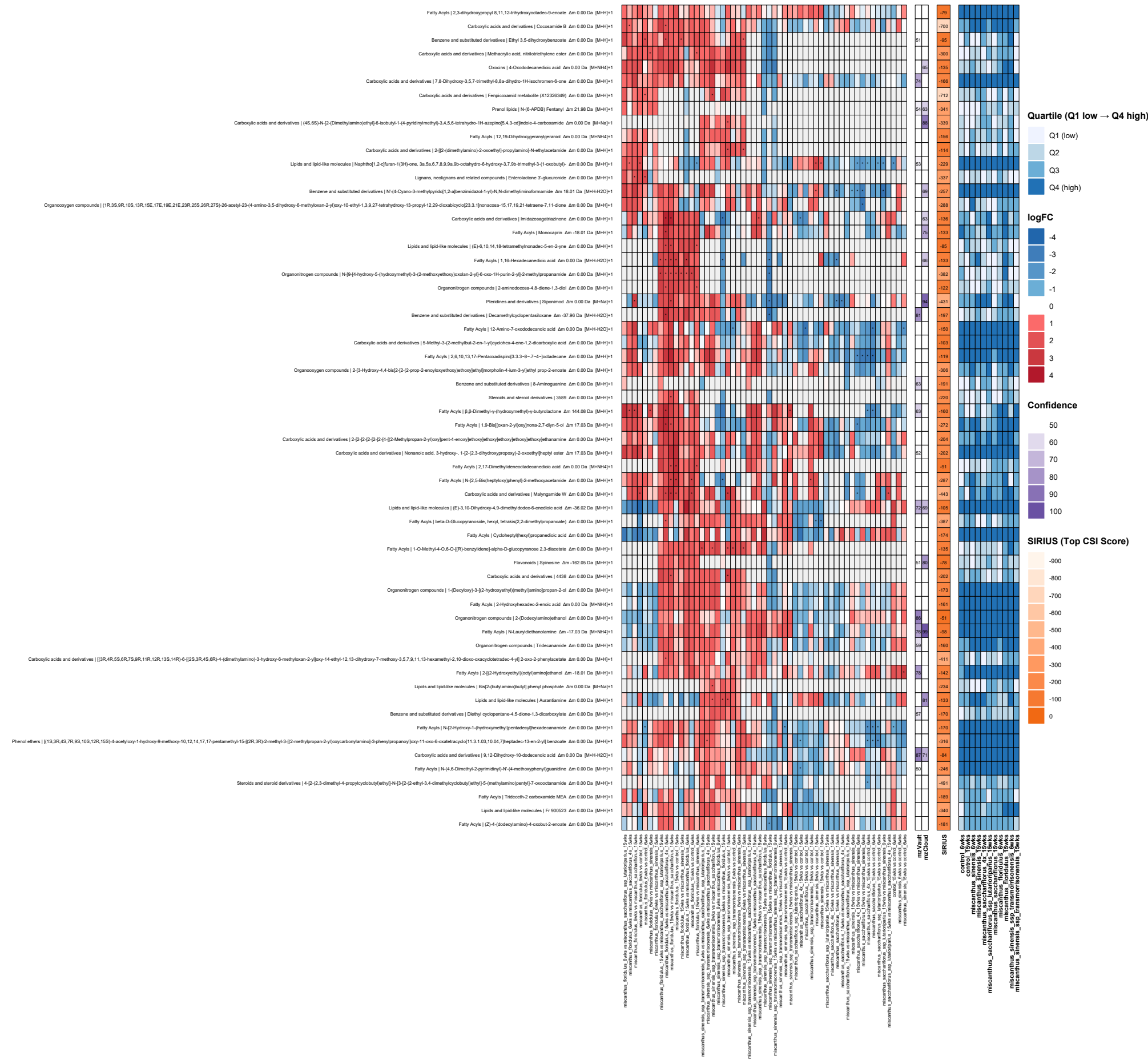
SIRIUS (Top CSI Score)

metabolite names and molecular weights/formulas (MW, MF) are listed on the left side of the heatmap. The columns are labeled with sample IDs (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13).



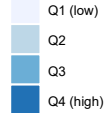
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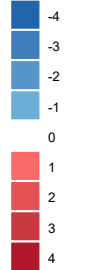


Fatty Acids N-[4E,8E]-1,3-dihydroxydocosa-4,8-dien-2-yl]propanamide Δm 0.00 Da [M+H] ⁺	
Fatty Acids Benzoylseramide Δm 0.00 Da [M+H] ⁺	
Carboxylic acids and derivatives [IR,45,55]-4-[[2R,3R]-3-[[2S]-1-(2S,5S,13S)-4,8-dimethyl-13-(methylamino)-3,14-dioxo-2-propan-2-yl]-1,4-diazacyclotetradecane-5-carboxyl]pyrrolidin-2-yl]-3-methoxy-2-methylpropanoyl]amino]-3-methoxy-5-methylheptanoic acid Δm 0.00 Da [M+H] ⁺	
Carboxylic acids and derivatives Methylsulfolipoyl C1 Amid Δm 0.00 Da [M+H] ⁺	
Fatty Acids 4-[[Ethyl]-2,5-dimethoxy-N-[2-methoxyphenyl]methyl]benzenesulfonamide Δm 0.00 Da [M+H] ⁺	
Fatty Acids 18,10,12,13-tetrahydroxyoctadecanoic acid Δm 0.00 Da [M+H] ⁺	
Organooxygen compounds [2]-2-(3-Hydroxy-4-(4-hydroxyphenyl)-5-oxotetra-2D(1H)-ylidene)acetic acid Δm -18.01 Da [M+H] ⁺	
Benzene and substituted derivatives Met-1-yl-Lys Δm 0.00 Da [M+H] ⁺	
Organic phosphoric acids and derivatives Tri(2-butylseryl) phosphate Δm 0.00 Da [M+H] ⁺	
Carboxylic acids and derivatives 2,2'-(mimosa(trimethylene)-6-(4,4-dimethyl-1,3-(2H,4H)isoquinoline)dione) Δm 0.00 Da [M+H] ⁺	
Carboxylic acids and derivatives (7E)-12-Hydroxy-3-isobutyl-13-methoxy-4,5,8-trimethyl-3,3a,4,6a,9,10,11,12,13,14-decahydro-1H-cyclooctadec[e]pindole-1,15(2H)-dione Δm 0.00 Da [M+H] ⁺	
Carboxylic acids and derivatives N-(2-Hydroxybutyl)-N-(2-hydroxyethyl)-beta-alanine Δm 0.00 Da [M+H] ⁺	
Lipids and lipid-like molecules methyl 1-(1-hexylpyridin-4-yl)-1H-1,2,3-benzoxazine-5-carboxylate Δm 18.01 Da [M+H+O] ⁺	
Carboxylic acids and derivatives 3,5-Di-tert-butyl-4-hydroxybenzoic acid Δm 0.00 Da [M+H] ⁺	
Organonitrogen compounds Farnesopropyl Δm 0.00 Da [M+H] ⁺	
Carboxylic acids and derivatives (E)-3,10-Dihydroxy-4,9-dimethyldeco-6-enedioic acid Δm 0.00 Da [M+H+O] ⁺	
Carboxylic acids and derivatives 2-(2-Methyl-1-[[2,2,4-trimethyl-6-(2-methylprop-2-en-1-yl)oxy]oxy]but-2-en-1-yl)-2-methylprop-2-enoate Δm 0.00 Da [M+H] ⁺	
Benzene and substituted derivatives N-Decamethylamadol Δm 0.00 Da [M+H] ⁺	
Fatty Acids 2-Propanone, 1,1,3,3-tetrapropyl Δm 0.00 Da [M+H] ⁺	
Phenol ethers 5-acetyl-1-(phenylsulfonyl)thiopyran Δm 0.00 Da [M+H] ⁺	
Fatty Acids Broussonetine E Δm 0.00 Da [M+H] ⁺	
Organooxygen compounds Neopentylene glycol Δm 0.00 Da [M+H] ⁺	
Allyl-type 1,3-dipolar organic compounds 2-Amino-N,N-dimethyladenosine Δm 0.00 Da [M+H] ⁺	
Organooxygen compounds N-(1,4-Dihydroxy-4-methylpentan-2-yl)-3-hydroxy-5-oxo-6-phenylhexanamide Δm -18.01 Da [M+H] ⁺	
Carboxylic acids and derivatives [Leu-Leu] Δm 0.00 Da [M+H] ⁺	
Carboxylic acids and derivatives 1-(6-methylcyclooctadecanoic acid Δm 0.00 Da [M+H] ⁺	
Fatty Acids 2-undecanemidoacetic acid Δm 0.00 Da [M+H] ⁺	
Phenol ethers butan-2-yl 7-(3,4-dimethoxyphenyl)-4-(3-hydroxy-4-methoxyphenyl)-2-methyl-5-oxo-4,6,7,8-tetrahydro-3H-quinoxaline-3-carboxylate Δm 0.00 Da [M+H] ⁺	
Carboxylic acids and derivatives Cyclo[neupyrtyl] Δm 0.00 Da [M+H] ⁺	
Fatty Acids (Z)-4-(2-(2-methoxyethoxy)ethoxy)ethoxybut-2-en-1-ol Δm 0.00 Da [M+H] ⁺	
Benzene and substituted derivatives Valproic Acid Δm 21.98 Da [M+H] ⁺	
Lipids and lipid-like molecules 5-Heptenoic acid, 7-[[1R,4S,5S,6R]-4-[[1E,3S)-3-hydroxy-1-oxopropyl]-2-oxopropyl]-2-[(hept-5-yl)-[5E]- Δm 0.00 Da [M+H] ⁺	
Fatty Acids 2,2'-[[4-(Tridecylphenyl)azaneyl]bis(methylene)]bis(prop-2-en-1-yl)acetic acid Δm 0.00 Da [M+H] ⁺	
Organooxygen compounds Neopentylene glycol Δm 0.00 Da [M+H] ⁺	
Carboxylic acids and derivatives [[2S,3S,5S,10S,13S,16S,17R]-3-acetyloxy-10,13-dimethyl-2-piperidin-1-yl]-1-prop-2-enyl]piperidin-1-um-1-yl]-2,3,4,5,6,7,8,9,11,12,14,15,16,17-tetradecahydro-1H-cyclopentad[e]phenanthren-17-yl]butanoate Δm 0.00 Da [M+H] ⁺	
Benzene and substituted derivatives O-Arachidonyl ethanolamine Δm 0.00 Da [M+H] ⁺	
Benzene and substituted derivatives Cindamycin Δm 0.00 Da [M+H] ⁺	
Carboxylic acids and derivatives Tetakis(vinyl)dimethylsiloxane Δm 0.00 Da [M+H] ⁺	
Organic phosphoric acids and derivatives Methyl [hydroxy(dimethoxy)benzo-6'-phosphoryl]dimethylphosphate Δm 0.00 Da [M+H] ⁺	
Carboxylic acids and derivatives 3-[4-amino-5-[4-[[benzene-2,4-di-5-(trifluoromethyl)phenyl]carbamoyl]amino]-3-fluorophenyl]pyrazole-2,1-T]-1,2-difluoro-7-yl]-N,N-dimethylpyridine-1-carboxamide Δm 0.00 Da [M+H] ⁺	
Organooxygen compounds 2-[[3R,4S]-3-[[5-(4-fluorophenyl)-1,2-oxazol-3-yl]methyl]-4-piperidinyl]-N-(4-methoxybenzoyl)acetamide Δm 21.98 Da [M+H] ⁺	
Benzene and substituted derivatives 3-[4-(Decoxyphenyl)-2-(ethoxymethyl)prop-2-en-1-yl]-2-oxo-1,3-dioxane Δm 0.00 Da [M+H] ⁺	
Phenol lipids 1,1,1-Trifluorooctane Δm 0.00 Da [M+H] ⁺	
Organooxygen compounds 1-Butoxy-3-(2-butylsethyl)propan-2-ol Δm 0.00 Da [M+H] ⁺	
Organonitrogen compounds n-Octadecylamine Δm 0.00 Da [M+H] ⁺	
Organonitrogen compounds 1-Heaudecylamine Δm 0.00 Da [M+H] ⁺	
Phenol ethers [5-[[1R,3aS,5aR,6aR,7aR,8S,11aR,11bR,13aR,13bR]-1-acetoxypentyl]-3a-(2-methoxyethyl)carbamoyl]-5a,9a,11a-pentamethyl-1,2,3,4,5,6,7,7a,9,10,11,11b,12,13,13a,13b-hexadecahydrocyclopentadec[gh]phenen-9-yl]oxy]-3,3-dimethyl-5-oxo-pentanoic acid Δm 0.00 Da [M+H] ⁺	
Organonitrogen compounds 2-[Tetradecylamino]ethanol Δm 0.00 Da [M+H] ⁺	
Coumarins and derivatives Barbitoside Δm -36.02 Da [M+H] ⁺	
Fatty Acids 6-O-Tetradecyl-D-galactose oxime Δm 0.00 Da [M+H] ⁺	
Fatty Acids Bis(4-vinylbutyl) succinate Δm 0.00 Da [M+H] ⁺	
Organooxygen compounds 2,2'-[2'-(1,3,5-Triazene-2,4,6-triazine)]bis(methanol Δm 0.00 Da [M+H] ⁺	
Benzene and substituted derivatives N-tert-Butyl-2-biopterinsulfonamide Δm 0.00 Da [M+H] ⁺	
Carboxylic acids and derivatives [dimethyl 2-[[1-(4,8-dioxo-1H-1,3,5-triazin-2-yl)-2-(2-methoxycyclohexa-1,3-dien-1-yl)amino]-2-oxoethyl]diazany]benzene-1,4-dicarboxylate Δm 0.00 Da [M+H] ⁺	
Organic phosphoric acids and derivatives ethyl [2E]-2-amino-2-[[2-(2-dimethoxyphosphoryl)ethoxy]imino]acetate Δm 0.00 Da [M+H] ⁺	
Carboxylic acids and derivatives 1-(2,2-Dimethoxyethyl)-4-nitrobenzene Δm 0.00 Da [M+H] ⁺	
Organooxygen compounds (E)-6-(6,6-trifluoro-2,2,5-trimethylhex-4-en-3-one Δm 0.00 Da [M+H] ⁺	
Carboxylic acids and derivatives Yo-17 Δm 0.00 Da [M+H] ⁺	
Fatty Acids cis-13,14-Epoxydocosanoic acid Δm 0.00 Da [M+H] ⁺	
Carboxylic acids and derivatives N-[2-(4-methoxymethyl)piperidin-4-yl]methyl-1-coucan-3-ylmethanamine Δm 0.00 Da [M+H] ⁺	

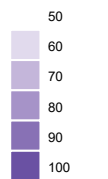
Quartile (Q1 low → Q4 high)



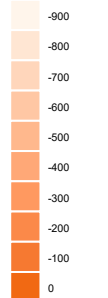
logFC



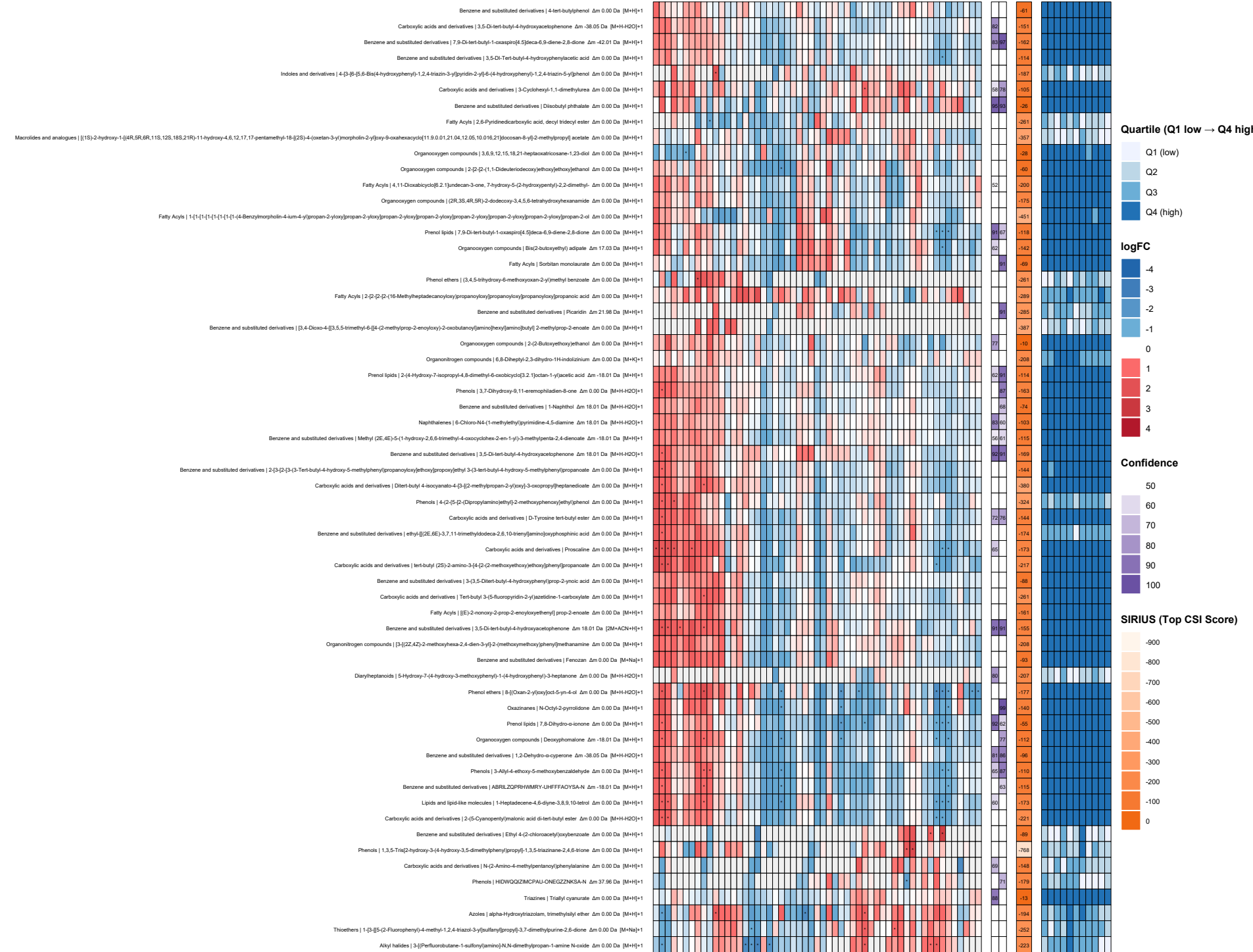
Confidence



SIRIUS (Top CSI Score)



miscanthus_1081	miscanthus_1082	miscanthus_1083	miscanthus_1084	miscanthus_1085	miscanthus_1086	miscanthus_1087	miscanthus_1088	miscanthus_1089	miscanthus_1090	miscanthus_1091	miscanthus_1092	miscanthus_1093	miscanthus_1094	miscanthus_1095	miscanthus_1096	miscanthus_1097	miscanthus_1098	miscanthus_1099	miscanthus_1100	miscanthus_1101	miscanthus_1102	miscanthus_1103	miscanthus_1104	miscanthus_1105	miscanthus_1106	miscanthus_1107	miscanthus_1108	miscanthus_1109	miscanthus_1110	miscanthus_1111	miscanthus_1112	miscanthus_1113	miscanthus_1114	miscanthus_1115	miscanthus_1116	miscanthus_1117	miscanthus_1118	miscanthus_1119	miscanthus_1120	miscanthus_1121	miscanthus_1122	miscanthus_1123	miscanthus_1124	miscanthus_1125	miscanthus_1126	miscanthus_1127	miscanthus_1128	miscanthus_1129	miscanthus_1130	miscanthus_1131	miscanthus_1132	miscanthus_1133	miscanthus_1134	miscanthus_1135	miscanthus_1136	miscanthus_1137	miscanthus_1138	miscanthus_1139	miscanthus_1140
miscanthus_1081	miscanthus_1082	miscanthus_1083	miscanthus_1084	miscanthus_1085	miscanthus_1086	miscanthus_1087	miscanthus_1088	miscanthus_1089	miscanthus_1090	miscanthus_1091	miscanthus_1092	miscanthus_1093	miscanthus_1094	miscanthus_1095	miscanthus_1096	miscanthus_1097	miscanthus_1098	miscanthus_1099	miscanthus_1100	miscanthus_1101	miscanthus_1102	miscanthus_1103	miscanthus_1104	miscanthus_1105	miscanthus_1106	miscanthus_1107	miscanthus_1108	miscanthus_1109	miscanthus_1110	miscanthus_1111	miscanthus_1112	miscanthus_1113	miscanthus_1114	miscanthus_1115	miscanthus_1116	miscanthus_1117	miscanthus_1118	miscanthus_1119	miscanthus_1120	miscanthus_1121	miscanthus_1122	miscanthus_1123	miscanthus_1124	miscanthus_1125	miscanthus_1126	miscanthus_1127	miscanthus_1128	miscanthus_1129	miscanthus_1130	miscanthus_1131	miscanthus_1132	miscanthus_1133	miscanthus_1134	miscanthus_1135	miscanthus_1136	miscanthus_1137	miscanthus_1138	miscanthus_1139	miscanthus_1140



Quartile (Q1 low → Q4 high)

logFC

Confidenc

SIRIUS (Top CSI Score)

